🛡️ 1. Introduction

TRINETRA is an advanced cyber defense platform designed to protect critical infrastructure against modern cyber threats. It integrates real-time monitoring, anomaly detection, and cross-system correlation to identify and respond to attacks proactively.

The system operates continuously, analyzing both external and internal signals to detect coordinated cyber campaigns.

🧠 2. Core Architecture

TRINETRA consists of three primary intelligence modules:

🔍 2.1 Drishti (Phishing Detection Engine)

- Monitors SSL certificates using real-time feeds (e.g., CertStream)
- Detects suspicious domains using:
  - Domain similarity scoring
  - Visual similarity analysis (UI cloning detection)
- Identifies phishing websites targeting government portals
- Provides takedown recommendations

🛡️ 2.2 Kavach (Anomaly Detection Engine)

- Detects abnormal system behavior using statistical models
- Monitors:
  - Login patterns
  - IP address deviations
  - Time-based anomalies (off-hours access)
  - Data transfer irregularities

📊 Z-Score Anomaly Detection

Z-score is used to measure how far a data point deviates from the mean.

Formula:

$Z = (x - \mu) / \sigma$

Where:

- x = observed value
- μ = mean
- σ = standard deviation

Interpretation:

- $|Z| < 2$ → Normal
- $|Z| \geq 2$ → Suspicious
- $|Z| \geq 3$ → Highly anomalous

Use Case:

- Detect unusual login spikes
- Identify abnormal user behavior
- Flag potential insider threats

🔗 2.3 Bridge (Correlation Engine)

- Links external threats with internal events
- Correlates:
  - Phishing domains → credential theft
  - Login anomalies → compromised accounts

Example:

1. Fake website detected (Drishti)
2. User enters credentials
3. Same credentials used internally (Kavach detects anomaly)
4. Bridge connects both → confirms attack chain

🌐 3. Threat Detection Capabilities

TRINETRA detects:

- Phishing attacks
- Domain spoofing
- Credential theft
- Unauthorized access
- Insider threats
- Coordinated cyber campaigns

📡 4. Real-Time Monitoring

The system continuously monitors:

- SSL certificate registrations
- Network traffic patterns
- Login activity across zones
- Endpoint behavior

🗺️ 5. Multi-Zone Protection

TRINETRA protects multiple infrastructure zones:

- Government portals
- Internal networks
- Authentication systems
- Critical databases

Each zone has:

- Independent anomaly detection
- Centralized correlation via Bridge

🚨 6. Alert System

TRINETRA generates alerts based on severity:

Severity	Description
LOW	Minor deviation
MEDIUM	Suspicious activity
HIGH	Confirmed anomaly
CRITICAL	Active attack detected

📊 7. Key Metrics (KPIs)

- Active threats detected
- Live phishing domains
- Login anomalies
- Cross-reality correlations
- Average detection time
- Zones protected

🔐 8. Security Features

- Real-time threat intelligence
- Behavioral analysis
- Automated alerting
- Cross-system correlation
- Adaptive learning models

🤖 9. AI Capabilities

TRINETRA uses machine learning to:

- Detect patterns in user behavior
- Classify phishing attempts
- Predict attack trends
- Reduce false positives

📜 10. Example Scenario

Attack Flow:

1. Attacker creates fake domain
2. Drishti detects domain similarity
3. User visits phishing site
4. Credentials stolen
5. Kavach detects unusual login
6. Bridge correlates both events
7. TRINETRA raises CRITICAL alert

📌 11. Conclusion

TRINETRA provides a unified cybersecurity solution by combining:

- Detection (Drishti)
- Analysis (Kavach)
- Correlation (Bridge)

This integrated approach enables early detection and rapid response to complex cyber threats.